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Printing plate picking system and process

Abstract

A printing plate picking mechanism includes a plate cassette stacking mechanism having an array of plate cassettes, each plate cassette adapted to hold a plurality of printing plates separated by intervening sheets of paper. A plate cassette movement mechanism is adapted to move a selected plate cassette to a printing plate loading position. A plate picker is adapted to contact and hold the leading edge of the topmost printing plate and transport it to plate transport pinch rollers. A paper picking unit includes a paper picker adapted to contact and hold a leading edge of a topmost sheet of paper in the selected plate cassette, and a plate guide positioned above the paper picker to guide the printing plate held by said plate picker. A vertical alignment system is adapted to vertically align the plate picker and the paper picking unit with the selected plate cassette.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention pertains to the field of printing plate loading systems and processes, and more particularly to the field of picking and conveying printing plates and interspersed slip sheets or paper from a plate cassette for conveyance to an imaging apparatus or system.

BACKGROUND OF THE INVENTION

(2) In the commercial printing industry, a step in the preparation of images for printing is the transfer of image information to an image recordable material that can be used repeatedly to print the image. While the image recordable material can take a variety of forms, one common form is a printing plate that includes a surface that can be modified in an image-wise fashion. Printing plates can take different forms. In one embodiment the modifiable surface includes a special coating referred to as an emulsion. An emulsion is a radiation sensitive coating that changes properties when exposed to radiation such as visible, ultraviolet, or infrared light. An emulsion can include one or more layers that are coated onto a substrate. The substrate can be composed of a variety of materials such as aluminum, polyester or elastomers.

(3) The transfer of image information to an image recordable material or the printing plate can be done in a variety of methods. One method in which image information is transferred to an image forming material is by computer-to-plate (CTP) systems. In CTP systems, images are formed on the modifiable surface of an image recordable material by way of radiation beams or the like generated by an imaging head in response to image forming information. In this manner, images can be quickly formed onto the image recordable material.

(4) It is advantageous to automate CTP systems to provide for shorter print runs and faster turn-around times. A CTP system can automate, through the use of computers and special equipment, the transfer of information from a digital file describing an original page layout to a printing plate.

In these types of CTP systems, unexposed printing plates are normally supplied in packages or cassettes in numbers that can range from a few dozen to several hundred, with slip-sheets (hereinafter referred to as paper or sheets of paper) interspersed between adjacent printing plates. The surfaces of these plates are delicate and sensitive and can be easily marred, which can result in undesirable defects in the final printed product. In order to protect the surfaces of these plates that are supplied in packages or cassettes, a sheet of paper is interspersed between the plates and provide a physical barrier between the plates. However, these sheets of paper need to be removed from between the printing plates prior to imaging. Therefore, any attempt to automate the handling of printing plates must include measures to prevent damage to the delicate modifiable surfaces of the plates and measures to also move the interspersed sheets of paper to a paper disposal unit in an automated fashion.
(5) The automation of paper removal and storage in this context presents a number of challenges. Paper removal is not simply a matter of moving a single sheet of paper from a stack of similar sheets of paper. In general, paper is made from materials different from those used for printing plates, and in particular is a material suitable for protecting and not damaging the modifiable surfaces of the printing plates. Separating a sheet of paper from an adjacent printing plate can be complicated when the sheet of paper becomes adhered to a surface of the adjacent printing plate by physical mechanisms that can include electrostatic attraction or the expulsion of air between the surfaces. These mechanisms can lead to multiple plate picks that can lead to system error conditions. Increasing plate-making throughput requirements complicate matters further by necessitating that the sheets of paper be removed at rates that do not hinder the increased plate supply demands.
(6) Conventional printing plate picking systems have typically picked and removed printing plates and interspersed sheets of paper sequentially from a media stack. For example, in some conventional systems, a sheet of paper is first picked from the media stack and moved to a disposal container. Once the sheet of paper has been moved, a printing plate is then picked and moved to a subsequent station where it is processed (e.g., by imaging in an exposure engine). In other conventional systems, a sheet of paper is picked and transferred to a disposal container after the printing plate has been secured and transferred to a subsequent process. In either case, the sequential picking and removal steps can adversely affect the overall system throughput times.
(7) Once a sheet of paper has been secured and separated from a printing plate, its reliable disposal presents additional challenges for automated media handling systems. Specifically, in a device designed to have a large number of printing plates on-line at any one time, the sheets of paper that are removed each time a plate is picked must be accumulated somewhere for disposal. Conventional plate-making systems have employed complex media handling mechanisms that remove and convey sheets of paper to containers such as slip-sheet or paper holders. The reliability and throughput of the media handling system may be adversely affected when a picked sheet of paper must be additionally conveyed and deposited into a slip-sheet holder. Further, when sheets of paper are crumpled during the act of picking, separating, conveying or depositing them into a slip-sheet or paper holder, the sheets of paper can occupy a significant volume that increases the size of the slip-sheet or paper holder, thus adversely impacting the required footprint of the plate-making system.
(8) Additionally, conventional plate loading and paper removal systems may require extensive floor space for accommodating a plate storage area, an imaging system and a transport space for moving a picked plate from the storage area to the imaging unit.
(9) FIG. **1** is a schematic representation of the elements or parts of a conventional printing plate handling system **10** for handling and imaging printing plates. The system **10** includes an imaging unit **1** (also referred to as a CTP device) where an image can be recorded on a printing plate. The imaging unit **1** can include an exposure system which is designed to record an image on the plate. In an automated arrangement, the printing plate handling system **10** can include an autoloader unit

3 that interfaces with a plate cassette array **5**. The plate cassette array **5** is designed to hold a stack of cassettes where each cassette in the stack of cassettes can each hold a plurality of plates that are to be picked and conveyed to the imaging unit **1** through the use of autoloader unit **3** to enable the imaging of the printing plate. In some cases, each cassette can hold a different type or size of plate to enable the automatic creation of different types of plates.

(10) FIG. **2** shows a known plate picking mechanism **20** of the type used in the KODAK MAGNUS VLF Platesetter Multi-Cassette Unit The plate picking mechanism **20** is adapted to load or convey plates **23** from a plate cassette **53***a* located in a plate cassette array **53** to an imaging unit **1** (FIG. **1**) such as a CTP device. FIG. **2** illustrates the interface between the plate picking mechanism **20** and the top or uppermost plate **23** in a plate stack **22** in a plate cassette **53***a*. As shown in FIG. **2**, the plate picking mechanism **20** includes a plate picking arrangement **27** that includes vacuum or lifting cups **29** mounted on a conveying arm **31** that is adapted to move the lifting cups **29** from a first position to a picking up position where the lifting cups **29** contact and pick or lift the uppermost plate **23** in the plate stack **22**. Once the uppermost plate **23** is lifted, the plate picking mechanism **20** further includes a plate support device **24** that is adapted to be moved to a position below the picked-up printing plate **23** as shown in FIG. **2**. The plate support device **24** includes an air suction arrangement that is designed to hold the plate **23** on an upper surface **24***a* of the plate support device **24** by way of the application of an air suction force on the lower surface of the plate **23**. The plate support device **24** also includes a paper sheet lifting and conveying mechanism **33** that is designed to lift a sheet of paper **35** that is interspersed between the lifted uppermost plate **23** and the next plate **23** in the plate stack **22** in the plate cassette **53***a*. In the arrangement of FIG. **2**, the plate support device **24** is adapted to convey the picked-up plate **23** over a plate support **26** to an imaging unit **1** (FIG. **1**) or CTP device, and is further adapted to convey the sheet of paper **35** to rollers **39** which direct the sheets of paper **35** into the disposal bin or paper disposal unit.

(11) In the arrangement of FIG. **2**, the plate cassette array is provided in a fixed vertical alignment or arrangement with a vertical gap **25** between the individual cassettes **53***a*, **53***b* in the plate cassette array **53**. The vertical gap **25** between adjacent plate cassettes **53***a*, **53***b* is sized and configured to permit the insertion of the plate picking arraignment **27** including the lifting cups **29**, conveying arm **31** and plate support device **24** therein to convey a plate **23** and sheet of paper **35**. This layout, which requires the handoff of the picked-up plate **23** to plate support device **24** and therefore includes the application of a vacuum force by separate elements along a plate transport path, also expands the footprint of the system, including the footprint of the system in a vertical direction.

(12) A further alternative known arrangement is the Agfa Advantage **3850** CL & CLS Cassette System. The Agfa **3850** system is an automatic flatbed violet laser Platesetter that enables access to the media stored within cassettes. The printing plates that are loaded in the Agfa **3850** system are sensitive to visible light, so the cassettes are designed to be light tight, and the cassettes are intended to be loaded in a separate room that is equipped with safelights. Once loaded with media the sealed cassette is moved to a trolley and docked to a CTP device. Inside the CTP device of the Agfa **3850** system a cassette support frame is positioned vertically by a scissor lift mechanism to receive the new cassette and roller guides are used to permit the cassettes to be automatically accessible within the CTP. When a specific media size is needed a cassette containing that media size is moved vertically to a picking height by the scissor lift mechanism. An air cylinder is used to horizontally locate the cassette at a picking position. In this system a mechanism guided by vertical slides is used to lower a front cover of the cassette to allow a picking mechanism to gain access to the media within. For removing media from the cassette, the Agfa **3850** system uses a stepper motor to horizontally move the picking mechanism along a central track and rod cylinders are used to vertically guide the picking mechanism. Suction cups mounted on the picking mechanism are used to pick the leading edge of the upper most interleaf paper and move the paper horizontally to a set of pinch rollers that pull the paper downwards into a bin. This same picking mechanism returns

to the cassette to pick the leading edge of the upper most plate using the same suction cups and moves the plate horizontally to capstan rollers which control the plate position during imaging on the flatbed platen. The Agfa **3850** system uses the same mechanism to sequentially move the interleaf paper and then the plate as opposed to moving both the interleaf paper and plate in a substantially parallel manner. This impacts the processing speed of the device and requires design compromises to enable a single mechanism to handle different media types (i.e., plates and paper).
(13) In an alternative known arrangement, the plates in a plate cassette array can be stacked in a fixed staggered arrangement that permits the leading edge of the plates and sheets of paper to be accessed by the plate picking arrangement. This staggered arrangement expands the horizontal footprint of the device.
(14) What is desired is a mechanism or arrangement having a reduced footprint that is capable of moving both interleaf paper and plates in a substantially parallel manner while protecting the plate from being damaged while it is conveyed.
SUMMARY OF THE INVENTION
(15) The present invention provides for an apparatus and process that is adapted to remove a printing plate from a plate stack in an array of plate cassettes that include printing plates and interspersed sheets of paper, wherein the apparatus is adapted to move a printing plate and sheet of paper in a substantially simultaneous manner to load the printing plate into an imaging system such as a CTP device, and to dispose the interspersed sheets of paper into a paper bin.
(16) In the apparatus of the present invention, an array of plate cassettes is arranged in a vertical configuration in a manner which minimizes the vertical spacing between adjacent stacked cassettes to thereby decrease a vertical footprint of the apparatus, and each of the cassettes in the stack is adapted to individually move horizontally with respect to the stack to provide for an opening that permits the insertion of a plate picker device and a paper picker device to convey plates and sheets of paper from the cassette.
(17) More specifically, the present invention relates to a printing plate picking mechanism, including: (i) plate transport pinch rollers adapted to direct a printing plate along a plate transport path; (ii) a paper disposal unit; (iii) a plate cassette stacking mechanism including: (a) an array of plate cassettes, each plate cassette adapted to hold a plurality of printing plates separated by intervening sheets of paper; and (b) a plate cassette movement mechanism adapted to move a selected plate cassette to a printing plate loading position thereby exposing a leading edge of a topmost printing plate in the selected plate cassette; (iv) a plate picker adapted to contact and hold the leading edge of the topmost printing plate and transport it to the plate transport pinch rollers; (v) a paper picking unit including: (a) a paper picker adapted to contact and hold a leading edge of a topmost sheet of paper in the selected plate cassette; and (b) a plate guide positioned above the paper picker, said plate guide forming a part of said plate transport path to guide the printing plate held by said plate picker to said plate transport pinch rollers; and (vi) a vertical alignment system adapted to move the array of plate cassettes or the paper picking unit in a vertical direction to vertically align the paper picking unit with the selected plate cassette.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) It is to be understood that the attached drawings are for purposes of illustrating the concepts of the invention and may not be to scale. Identical reference numerals have been used, where possible, to designate identical features that are common to the figures.
(2) FIG. **1** is a schematic representation of a conventional CTP system;
(3) FIG. **2** shows a known plate picking mechanism;
(4) FIG. **3** shows an exemplary printing plate picking mechanism;

(5) FIGS. **4**A-**4**B show an exemplary plate picker configuration;
(6) FIGS. **4**C-**4**F show an exemplars paper picker configuration;
(7) FIGS. **5**A-**5**E illustrate the operation of the exemplary plate picker and paper picker assembly; and
(8) FIG. **6** shows a flow chart of the operation of the printing plate picking mechanism in accordance with an exemplary embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(9) The invention is inclusive of combinations of the embodiments described herein. References to "a particular embodiment" and the like refer to features that are present in at least one embodiment of the invention. Separate references to "an embodiment" or "particular embodiments" or the like do not necessarily refer to the same embodiment or embodiments; however, such embodiments are not mutually exclusive, unless so indicated, or as are readily apparent to one of skill in the art. The use of singular or plural in referring to the "method" or "methods" and the like is not limiting. It should be noted that, unless otherwise explicitly noted or required by context, the word "or" is used in this disclosure in a non-exclusive sense.

(10) FIG. **3** illustrates an exemplary printing plate picking mechanism **50** including a plate picker and paper picker mechanism **70** (also described herein as a plate picker and paper picker unit, assembly or arrangement) in accordance with a feature of the present invention. The printing plate picking mechanism **50** can be part of an overall system as schematically illustrated in FIG. **1**. As shown in FIG. **3**, the printing plate picking mechanism **50** includes a plate cassette stacking mechanism or plate cassette array **53** which comprises a stack a plate cassettes **53***a*-**53***f* which are each configured to hold a plurality a printing plates **55** with interspersed or intervening sheets of paper. As shown in FIG. **3**, the plate cassettes **53***a*-**53***f* in the plate cassette array **53** are vertically aligned in a stack in a manner which minimizes a spacing **57** between adjacent plate cassettes **53***a*-**53***f* and therefore minimizes the overall size of the plate cassette array **53** in the vertical direction.

(11) In accordance with the present invention, a selected plate cassette **100** (plate cassette **53***b* in FIG. **3**) can be conveyed from a home position where all of the plate cassettes **53***a*-**53***f* in the paper cassette array **53** are horizontally aligned (for example, reference is made to the horizontally aligned position of each plate cassette **53***a* and **53***c*-**53***f* in FIG. **3** which are all in the same horizontal position) to a printing plate loading position **61**. This is achieved by moving the selected plate cassette **100** to the cassette loading position **61** in direction H. FIG. **3** shows the selected plate cassette **100** (i.e., plate cassette **53***b*) after movement of the selected plate cassette **100** in direction H to the loading position **61**. When the selected plate cassette **100** is moved to the printing plate loading position **61**, an exposed portion **100***a* of a topmost printing plate **55** (including the leading edge of the printing plate **55**) is exposed in the cassette **53***b*.

(12) The movement of the selected plate cassette **100** can be achieved by any known type of plate cassette movement mechanism **145** that is adapted to move the selected plate cassette **100** in horizontal direction H to position the selected plate cassette **100** in the loading position **61**. In some embodiments, the plate cassette movement mechanism **145** can include and be adapted to use one or more motors to drive wheels attached to the selected plate cassette **100** or any other type of lever, gear, hydraulics, rack and pinion, cables and pulleys, etc., that is adapted to move the selected plate cassette **100** in a horizontal manner in direction H. As a further option, a selected plate cassette **100** can be adapted to be manually pushed or pulled to the loading position **61**.

(13) In a further feature of the invention, a vertical alignment system **140** can be used to move the plate cassettes **53***a*-**53***f* in the plate cassette array **53** vertically through any known driving means for achieving vertical movement. For example, the cassettes can be operationally associated with a drive screw or a belt drive or the like which can be driven by a motor to cause a corresponding vertical movement of the plate cassette array **53** to vertically align the selected plate cassette **100** with the loading position **61**. Alternatively, the vertical alignment system **140** can be adapted to move the plate picker and the paper picking unit **70** in the vertical direction V to vertically align the

selected plate cassette **100** with the plate picker and paper picker unit **70**.

(14) As described with reference to FIG. **1**, an auto-loader unit **3** is adapted to interface with the plate cassette array **5**. With reference to FIG. **3**, an auto loader unit in accordance with the present invention includes the plate picker and paper picker mechanism **70** which interfaces with the plate cassette array **53**. In an automated CTP system, a printing plate **55** can be moved from a selected plate cassette **100** in an automated matter so as to be conveyed to an imaging unit **1** (FIG. **1**), while in a substantially parallel manner an interspersed sheet of paper can also be conveyed in an automated matter to a paper bin or paper disposal unit **71**.

(15) As noted, relative to FIG. **3**, in an embodiment of the present invention, the plate cassettes **53***a*-**53***f* can be moved in a vertical direction by a known mechanism adapted to move the plate cassettes **53***a*-**53***f* to permit the movement of the selected plate cassette **100** (e.g., plate cassette **53***b*) to a loading position **61**. However, the present invention is not limited thereto, and as an alternative arrangement, elements of the plate picker and paper picker mechanism **70** can also be adapted to move in a vertical direction to align with a selected plate cassette **100** (e.g., plate cassette **53***b*) for moving a top-most printing plate **55** within the selected plate cassette **100** to the imaging unit **1**.

(16) The plate picker and paper picker mechanism **70** is adapted to convey a selected printing plate **55** to an imaging device through a plate conveyor **74**. The conveyor **74** is adapted to convey a printing plate **55** to a CTP device for imaging. Additionally, with the arrangement of the present invention, the plate picker and paper picker mechanism **70** is further adapted to convey paper to a paper bin or paper disposal unit **71**. The plate picker and paper picker mechanism **70** and specifically the plate picker and paper picker operations of the plate picker and paper picker mechanism **70** can work separately and/or as a system. Each of the plate picking and paper picking functions of the plate picker and paper picker mechanism **70** can be a stand-alone unit combined to perform a plate or paper movement from a plate cassette **53***a*-**53***f* (printing plate **55** to conveyor **74**, and paper to paper disposal unit **71**). In a representative example of the present invention, each of the plate picker and paper picker of the plate picker and paper picker mechanism **70** can handle a 50 mm stack of printing plates **55** and paper from a selected plate cassette **100** selected from the set of plate cassette **53***a*-**53***f* Additionally, each working plate cassette **53***a*-**53***f* can be moved (pushed or pulled by a mechanism) to loading position **61** in order to expose and reveal an exposed potion **100***a*. The exposed portion **100***a* can also be referred to as an operation zone.

(17) Referring to FIGS. **4**A and **4**B, a plate picker **70***a* which is part of the plate picker and paper picker mechanism **70** (FIG. **3**) is illustrated in isolation. The plate picker **70***a* includes a plurality of vacuum cups **73** (also referred to as suction cups) mounted on an assembly having a horizontal bar or rail **77** and vertical bars or rails **77***a* and **77***b*. The vacuum cups **73** are adapted to apply a suction or vacuum force onto the plate to lift the plate, and is supplied to the vacuum cups **73** by way of ally known vacuum source. The vacuum cups **73** are mounted on rails **77***a*, **77***b* so as to be movable in vertical directions A, B by way of a DC motor **75** and a timing or drive belt **76**. The vacuum cups **73** are also movable in horizontal directions C, D along horizontal rails **83** by way of a DC motor **80** and a timing or drive belt **81**. In accordance with an embodiment of the invention, the motor **80** and drive belt **81** are operationally associated with vertical rails **77***a*, **77***b* of the assembly upon which the vacuum cups **73** are mounted, such that a driving movement applied by the motor **80** and drive belt **81** causes a corresponding movement of the assembly and thereby the mounted vacuum cups **73** in horizontal directions C, D. In operation, and as will be described in detail, for picking a plate in accordance with the present invention, the vacuum cups **73** are moved through the use of motors **75** and **80** to position the vacuum cups over a printing plate located in a plate cassette by driving the vacuum cups in direction C or D through motor **80** to position the vacuum cups over the plate, and moving the vacuum cups downward in direction B by operation of motor **75** to cause the vacuum cups **73** to contact and hold a selected printing plate. A further tilting motion of the assembly and specifically vacuum cups **73** can be achieved through a DC tilt motor **82** which is adapted to provide a tilting or angular movement of the vacuum cups **73** relative to the

bars **77**, **77***a* and **77***b*. This can aid in the separation of a picked printing plate **105** from an underlying sheet of paper **400** (e.g., see FIG. **5**C) by tilting the vacuum cups **73** relative to a horizontal plane. Air blowers **72** can also be activated to aid in the separation. In some embodiments, the entire assembly can alternatively be tilted.

(18) While the embodiment illustrated in FIG. **4**A shows the use of motors and drive belts to achieve the desired movement of the vacuum cups **73**, the present invention is not limited thereto. It is recognized that any type of device that is adapted to provide vertical, horizontal and tilting movements such as hydraulics, levers, gears, and rack and pinion arrangements can be utilized within the context of the present invention.

(19) Accordingly, the plate picker **70***a* can be a bar with vacuum cups that attaches and/or is adapted to contact a printing plate **105** and moves the printing plate **105** along a plate transport path from a selected plate cassette **100** to plate transport pinch rollers **59** (see FIG. **4**B). The plate transport pinch rollers **59** direct the plate **105** to conveyor **74** (also known as a conveying bridge) to convey the printing plate into an imaging device or CTP device. As noted above, the vacuum cups **73** of plate picker **70***a* can move horizontally, vertically, and can have a tilted (pendulum) movement. The horizontal, vertical and tilting movements can be generated by the illustrated DC motors (**75**, **80**, **82**) attached to pulleys that are connected to drive belts **76**, **81**. The present invention is not limited thereto, and the DC motors can be connected to any belt type, rope or cable. The timing belts can be connected to a bracket and transfer the forces generated from the DC motor to move the plate picker structure. These movements can also be generated by one or more cylinders (hydraulic and/or pneumatic). The arm can be tilted by a DC motor **82** connected to a plate. As also described, the horizontal and vertical movements can be guided by rails, and in a feature of the invention, these movements can be detected by sensors at each relevant position which provide a signal to a control device which is adapted to control the operation of the plate picker and paper picker mechanism **70**.

(20) FIG. **4**C illustrates in isolation a paper picking unit that includes a paper picker **70***b* which is part of the plate picker and paper picker mechanism **70** (FIG. **3**). FIG. **4**D shows the paper picker **70***b* in the context of other system components. As shown in FIG. **4**C, the paper picker **70***b* includes a plurality of suction or vacuum cups **170** mounted on a bar or rail **172** which makes up part of the paper picker **70***b* or paper picker assembly. The vacuum cups **170** are adapted to apply a suction or vacuum force that is applied onto the paper to lift the paper and is supplied to the vacuum cups **170** by way of any type of known vacuum source. The paper picker **70***b* also includes guide rollers **300** which function as a plate guide to guide a printing plate **105** over the top of the paper picker **70***b*. In other configurations, other types of plate guides can be used instead of guide rollers **300**. For example, the plate guides can be a fixed smooth surface that the printing plate **105** can slide over.

(21) The paper picker **70***b* with the rail **172** and the vacuum cups **170** attached thereto is adapted to move vertically through the use of vertical actuators **174** or a motor and drive belt mechanism connected thereto to move the rail **172** vertically. The present invention is not limited to the use of an actuator or a motor and drive belt, and any other type of known driving mechanism adapted to move the bar in the vertical direction can be utilized within the context of the present invention. The paper picker **70***b* can also move horizontally and includes a pair of guide bars or rails **176** which are adapted to guide the paper picker **70***b* in a horizontal direction when driven by an associated drive motor **177** and drive belt **175**. For example, the drive motor can be adapted to drive a drive belt attached to a wheel for moving the paper picker **70***b* and specifically the vacuum cups **170** in a horizontal direction along the rails **176**. The path that the paper takes is schematically illustrated by paper path shown in FIGS. **4**E and **4**F.

(22) The vacuum cups **170** of the paper picker **70***b* can be positioned over a sheet of paper **400** in the selected plate cassette **100** and activated to lift the sheet of paper **400**. The vacuum cups **170** can then be pivoted (e.g., by moving a roller **601** along a cam surface **600**) to pinch the sheet of

paper **400** between the vacuum cups **170** and a paper stopper **171**. This enables the paper **400** to be pulled to a dispose position over pinch rollers **178***a* as shown in FIG. **4**E. When the paper picker **70***b* is at the dispose position, the pinch rollers **178***a* are opposed to and are operationally associated with a pair of rollers **178***b*.
(23) As more clearly shown in FIG. **4**F, when the paper picker **70***b* is at the dispose position, the paper **400** is positioned between the pinch rollers **178***a* and rollers **178***b*. The vacuum cups **170** can be pivoted to un-pinch the paper **400** and the pinch rollers **178***a* can be activated to draw the paper **400** between the pinch roller **178***a* and direct the paper **400** to a sloped surface **180** which directs the paper **400** into a paper bin or paper disposal unit **71** (FIG. **3**).
(24) The paper picker **70***b* is essentially a bar with vacuum cups **170** that attaches to a piece of paper **400** that is inserted between adjacent printing plates **105**, which tweezes it and moves it from a cassette towards pinch rollers **178***a*. The pinch rollers **178***a* cooperate with rollers **178***b* and are adapted to pull the paper **400** down so that it slides upon sloped surface **180** into a paper bin.
(25) As noted above, the paper picker **70***b* moves horizontally and vertically. In an exemplary configuration, the horizontal movement is generated by a DC motor attached to a pully that is connected to a drive belt. However, the present invention is not limited thereto, and the DC motor can be attached to any belt type, rope or cable. The timing belt can be connected to a bracket and transfers the forces generated from the motor to move the paper picker **70***b*. These movements can also be generated by a cylinder (hydraulic and/or pneumatic). The vertical movement can be generated by an actuator **174** such as an air cylinder or hydraulic cylinder or DC motor. As shown, the horizontal and vertical movement can be guided by rails. Alternatively, any mechanism known in the art can be used to move and guide the paper picker **70***b*. As will be described with reference to FIGS. **5**A-**5**E, the movements described above can be detected by sensors at each relevant position which can be adapted to provide a signal to a control device for controlling the operation of the overall device including the plate picker **70***a* and the paper picker **70***b*.
(26) Operation of the plate picking and paper picking mechanism will now be described with reference to FIGS. **5**A-**5**E as well as FIGS. **4**A-**4**E. Referring first to FIGS. **5**A-**5**E, these figures illustrate a sequence for picking up a plate **105** and interspersed piece of paper **400**, and the interaction between the plate picker **70***a* and the paper picker **70***b*. As shown in FIG. **5**A, a selected plate cassette **100** having a plurality of plates and interspersed paper is first moved to an operating or loading position **61** (for example, as shown in FIG. **3**). It is noted that the selected plate cassette **100** is part of a plate cassette array **53**, wherein the cassettes **53***a*-**53***f* in the plate cassette array **53** are vertically stacked in a manner which minimizes the vertical footprint of the plate cassette array **53** as illustrated in FIG. **3**. As described with reference to FIG. **3**, and also as shown in FIG. **5**A, the selected plate cassette **100** is adapted to move to loading position **61** to reveal an exposed portion **100***a*. The vacuum cups **73** of the plate picker **70***a* are then adapted to move in horizontal and vertical directions and to provide a tilting movement as also described with reference to FIGS. **4**A and **4**B to pick up a top-most printing plate **105** as shown in FIG. **5**A. Air blowers **72** can optionally be used to aid in the separation of the printing plate **105** from the underlying paper **400**.
(27) Once the printing plate **105** is lifted to the position shown in FIG. **5**A, the top surface of an intervening paper **400** is exposed in the exposed portion **100***a* of the selected plate cassette **100** to permit the insertion of the paper picker **70***b* below the lifted printing plate **105**. With respect to the plate picker **70***a*, horizontal motor **80** (FIG. **4**A) can be activated to move the vacuum cups **73** and the printing plate **105** attached thereto along a plate transport path over guide rollers **300** to direct the plate **105** in a horizontal direction to the position shown in FIG. **5**B. In an exemplary embodiment, the guide rollers **300** are positioned above the paper picker **70***b*, preferably as part of the same structure. In other embodiments other types of paper guides besides guide rollers **300** can be used as will be obvious to one skilled in the art. Furthermore, the guide rollers **300** or paper guide can alternately be attached to some other structure besides the structure of the paper picker **70***b*.

(28) As shown in FIG. **5**B, the plate picker **70***a* with vacuum cups **73** convey the printing plate **105** along the plate transport path to pinch rollers **59** which guide the printing plate **105** to an imaging or CTP device via the conveyor **74**. As noted, sensors associated with the position of the plate picker **70***a* can cooperate with further sensors along the plate transport path and can be adapted to provide signals to a plate picker motion control system **159** (also referred to as a control device) indicative of the position of the picked-up printing plate **105** and/or the plate picker **70***a* to permit and/or facilitate an automated control of the process based on the position of the printing plate **105** and/or the plate picker **70***a*. The plate picker motion control system **159** controls the motion of the plate picker **70***a* according to a specified plate picker motion pattern by controlling motion control components such as motors. In a similar manner the plate picker motion control system **159** can be adapted to receive feedback and/or signals from the motors and drive belts associated with the movement of the plate picker **70***a* or different elements of the plate picker **70***a* to facilitate and/or control operation of the system.
(29) As shown in FIG. **5**C, as the printing plate **105** is in the process of being picked up and conveyed, the paper picker **70***b* is moved to an operative position which permits the vacuum cups **170** to attach to and pick up the paper **400** below the picked-up printing plate **105** and convey the paper **400** to the pinch rollers **178***a*. As discussed with reference to FIGS. **4**C and **4**D, in an exemplary configuration the movement of the paper picker **70***b* and paper vacuum cups **170** in the horizontal direction is achieved through the use of the actuator and motors. The tilting movement of the paper vacuum cups **170** can be achieved through the combination of a roller **601** which follows a cam surface **600** when driven by a known driving mechanism such as a motor to move the vacuum cups between a substantially horizontal position as shown in FIG. **4**F and a tilted position as shown in FIG. **4**E.
(30) As illustrated in FIG. **5**D, the paper picker **70***b* is adapted to move the paper **400** to a position where the pinch rollers **178***b* oppose the pinch rollers **178***a* and the paper **400** is attached to the vacuum cups **170**. Tilting of the paper picker **70***b* and the vacuum cups **170** through the roller **601** and cam surface **600** place the rollers **178***b* in a tangent or opposing position with respect to the pinch rollers **178***a* to ensure that the paper is driven downward direction (see FIG. **5**E and FIG. **4**F). Additionally, the vacuum cups **170** are operated through the use of the motors and actuators as shown in FIGS. **4**C and **4**D to pull the paper from the cassette while the pinch rollers **178***a* are operated to direct the paper **400** downward through the pinch rollers **178***a* directly to a paper disposal unit **71** (FIG. **3**) or alternatively to sloped surface **180** that directs the paper to a paper disposal unit **71**.
(31) As described with respect to the plate picker **70***a*, sensors are provided along the paper transport path or are associated with the paper picker **70***b* to detect the position of the paper **400** and/or paper picker **70***b* and provide a signal to a paper picker motion control system **590** that is adapted to control a motion of the paper picker **70***b* according to a specified motion pattern to enable the paper **400** to follow the paper transport path as shown in FIGS. **5**A-**5**E and FIGS. **4**D-**4**E. More specifically, sensors associated with the position of the paper picker **70***b* can cooperate with further sensors along the paper transport path and be adapted to provide a signal to the paper picker motion control system **590** indicative of the position of the paper **400** and/or the paper picker **70***b* to permit and/or facilitate an automated control of the process based on the position of the printing plate **105**, the plate picker **70***a*, the paper **400** and/or paper picker **70***b*. The paper picker motion control system **590** controls the motion of the paper picker **70***b* according to a specified paper picker motion pattern by controlling motion control components such as motors. In a similar manner, the paper picker motion control system **590** can be adapted to receive feedback and/or signals from the motors and drive belts associated with the movement of the paper picker **70***b* or different elements of the paper picker **70***b* to facilitate and/or control operation of the system.
(32) FIG. **5**C illustrates the picked-up printing plate **105** being moved toward the pair of pinch

rollers **59** through the use of the plate picker **70***a* and the paper picker **70***b* being positioned to begin the movement of the interspersed paper **400** located below the picked-up printed plate **105** from the cassette **100**. As the printing plate **105** is conveyed through the use of the pinch rollers **59** into an imaging or CTP device as shown in FIG. **5**D, the paper **400** is positioned so as to be held by the vacuum cups **170** and located between the rollers **178***b* and the pinch rollers **178***a* as shown. FIG. **5**E (as well as FIG. **4**F) show the paper **400** being fed through the pinch roller **178***a* toward the sloped surface **180** to be directed into paper disposal unit **71** (FIG. **3**). As shown in FIG. **4**F, in order to permit the paper **400** to be driven by the pinch rollers **178***a* into the paper disposal unit **71**, the vacuum cups **170** are controlled to release the paper **400**.

(33) In a feature of the present invention, the plate picker motion control system **159** and paper picker motion control system **590** are adapted to work together in an integrated manner or separately to control an operation of the system. FIG. **6** illustrates a flow chart showing an operation of the system and process in accordance with a feature of the invention based on the plate picker motion control system **159** and the paper picker motion control system **590**. As shown in FIG. **6**, a first step comprises initiating an initiate plate picking operation step **2000**. Thereafter, in a control vertical alignment of plate picker step **2001**, the plate picker motion control system **159** is adapted to proceed with the operation by controlling a vertical alignment of plate cassette array **53** (FIG. **3**) such that a selected plate cassette **100** in the set of plate cassettes **53***a*-**53***f* is aligned with the plate picker **70***a*.

(34) A control horizontal movement of plate cassette step **2002** controls the movement of the selected plate cassette **100** in the horizontal direction to move the plate cassette **100** into the plate loading position **61** to provide an exposed portion **100***a*, which exposes the leading edge of the topmost printing plate **105** in the selected plate cassette **100**.

(35) In a contact leading edge of printing plate step **2003**, the plate picker **70***a* is controlled to contact and hold the leading edge of the topmost printing plate **105** and lift it in a vertical direction exposing the leading edge of the topmost sheet of paper **400** in the selected paper cassette **100** (see FIG. **5**A).

(36) In a move paper picker over exposed leading edge of paper step **2004**, the paper picker **70***b* is controlled to move the paper picker **70***b* over the exposed leading edge of the topmost sheet of paper **400** in the selected paper cassette **100** (see FIG. **5**B).

(37) In a transport printing plate to pinch rollers step **2005**, the plate picker **70***a* is controlled by the plate picker motion control system **159** to transport the topmost printing plate **105** over the plate guide (e.g., guide rollers **300**) of the paper picker **70***b* and feed the leading edge of the topmost printing plate **105** into the plate transport pinch rollers **59** (see FIGS. **5**C-**5**D).

(38) In a transport printing plate step **2006**, the plate transport pinch rollers **59** are controlled to direct the topmost printing plate **105** along the plate transport path toward the imaging unit **1** (FIG. **1**) via the conveyor **74**. In a grab leading edge of paper step **2007**, the paper picker **70***b* is controlled to grab the leading edge of the topmost sheet of paper in the selected plate cassette **100** (FIGS. **5**B-**5**C).

(39) In a direct paper to disposal step **2008**, the topmost sheet of paper **400** is transported along a paper transport path and the topmost sheet of paper **400** is directed into the paper disposal unit **71** (see FIGS. **5**D-**5**E).

(40) It should be noted that the steps controlling the motion of the printing plate **105** (e.g., steps **2003**, **2005**, **2006**) are generally done in parallel with the steps controlling the motion of the paper **400** (e.g., steps **2004**, **2007**, **2008**). For example, the transport printing plate to pinch rollers step **2005** will preferably be performed simultaneously with the direct paper to disposal step **2008** such that the printing plate **105** and the sheet of paper **400** are both being moved together. This enables the process to be faster than if the steps were performed sequentially and further enables the protection of the underlying plate or next plate in the stack to be picked by assuring that paper is always positioned between the picked up plate **105** and the next plate **105** in the stack.

(41) The invention has been described in detail with particular reference to certain preferred embodiments thereof, but it will be understood that variations, combinations, and modifications can be affected by a person of ordinary skill in the art within the spirit and scope of the invention.

PARTS LIST

(42) **1** imaging unit **3** autoloader unit **5** plate cassette array **10** printing plate handling system **20** plate picking mechanism **22** plate stack **23** plate **24** plate support device **24***a* upper surface **25** vertical gap **26** plate support **27** plate picking arrangement **29** lifting cups **31** conveying arm **33** paper lifting and conveying mechanism **35** sheet of paper **39** rollers **50** plate picking mechanism **53** plate cassette array **53***a* plate cassette **53***b* plate cassette **53***c* plate cassette **53***d* plate cassette **53***e* plate cassette **53***f* plate cassette **55** printing plate **57** spacing **59** pinch rollers **61** loading position **70** plate picker and paper picker mechanism **70***a* plate picker **70***b* paper picker **71** paper disposal unit **72** blower **73** vacuum cups **74** conveyor **75** motor **76** drive belt **77** rail **77***a* rail **77***b* rail **80** motor **81** drive belt **82** motor **83** rail **100** selected plate cassette **100***a* exposed portion **105** printing plate **140** vertical alignment system **145** plate cassette movement mechanism **159** plate picker motion control system **170** vacuum cups **171** paper stopper **172** rail **174** actuator **175** drive belt **176** rail **177** motor **178***a* pinch roller **178***b* rollers **180** sloped surface **300** guide rollers **400** paper **590** paper picker motion control system **600** cam surface **601** roller **2000** initiate plate picking operation step **2001** control vertical alignment of plate picker step **2002** control horizontal movement of plate cassette step **2003** contact leading edge of printing plate step **2004** move paper picker over exposed leading edge of paper step **2005** transport printing plate to pinch rollers step **2006** transport printing plate step **2007** grab leading edge of paper step **2008** direct paper to disposal step

Claims

1. A printing plate picking mechanism, comprising: (i) plate transport pinch rollers adapted to direct a printing plate along a plate transport path; (ii) a paper disposal unit; (iii) a plate cassette stacking mechanism including: (a) an array of plate cassettes, each plate cassette adapted to hold a plurality of printing plates separated by intervening sheets of paper, and (b) a plate cassette movement mechanism adapted to move a selected plate cassette to a printing plate loading position thereby exposing a leading edge of a topmost printing plate in the selected plate cassette; (iv) a plate picker adapted to contact and hold the leading edge of the topmost printing plate and transport it to the plate transport pinch rollers; (v) a paper picking unit including: (a) a paper picker adapted to contact and hold a leading edge of a topmost sheet of paper in the selected plate cassette and transport said topmost sheet of paper to the paper disposal unit as said topmost printing plate held by said plate picker is transported to said plate transport pinch rollers, wherein said topmost sheet of paper and topmost printing plate are transported substantially simultaneously; and (b) a plate guide positioned above the paper picker, said plate guide forming a part of said plate transport path to guide the printing plate held by said plate picker to said plate transport pinch rollers; and (vi) a vertical alignment system adapted to move the array of plate cassettes or the plate picker and the paper picking unit in a vertical direction to vertically align the plate picker and the paper picking unit with the selected plate cassette.
2. The printing plate picking mechanism according to claim 1, further comprising: (vii) a paper picker motion control system adapted to control motion of the paper picking unit according to a specified motion pattern; and (viii) a control system adapted to: control the vertical alignment system to vertically align the plate picker and the paper picking unit with the selected plate cassette; control the plate cassette movement mechanism to move the selected plate cassette in the horizontal direction to the plate loading position to expose the leading edge of the topmost printing plate in the selected plate cassette; control the plate picker to contact and hold the leading edge of the topmost printing plate and lift it in a vertical direction exposing the leading edge of the topmost sheet of paper in the selected paper cassette; control the paper picker motion control system to

move the paper picking unit over the exposed leading edge of the topmost sheet of paper in the selected paper cassette; control the plate picker to transport the topmost printing plate over the plate guide of the paper picking unit and feed the leading edge of the topmost printing plate into the plate transport pinch rollers; control the plate transport pinch rollers to direct the topmost printing plate along the plate transport path; control the paper picker to grab the leading edge of the topmost sheet of paper in the selected plate cassette; and control the paper picker motion control system to transport the topmost sheet of paper along a paper transport path and direct the leading edge of the topmost sheet of paper into the paper disposal unit.

3. The printing plate mechanism according to claim 1, wherein the plate picker includes a plurality of vacuum cups adapted to apply a vacuum force onto the plate to lift the plate.

4. The printing plate mechanism according to claim 1, wherein the plate guide includes guide rollers.

5. The printing plate mechanism according to claim 1, wherein the plate picker includes air blowers adapted to facilitate a separation between a lifted plate and an adjacent sheet of paper.

6. The printing plate mechanism according to claim 1, wherein the plate cassettes in said array of plate cassettes are aligned in a stack and said plate cassette movement mechanism is adapted to move the selected plate cassette in a horizontal direction to said printing plate loading position to expose the leading edge of the topmost plate.

7. The printing plate mechanism according to claim 1, wherein said paper picking unit further includes at least one pair of opposing rollers, said plate picker being adapted to transport a sheet of paper to said at least one pair of opposing rollers and said at least one pair of opposing rollers being adapted to pull the sheet of paper into said paper disposal unit.

8. A printing plate picking mechanism, comprising: (i) plate transport pinch rollers adapted to direct a printing plate along a plate transport path; (ii) a paper disposal unit; (iii) a plate cassette stacking mechanism including: (a) an array of plate cassettes, each plate cassette adapted to hold a plurality of printing plates separated by intervening sheets of paper; and (b) a plate cassette movement mechanism adapted to move a selected plate cassette to a printing plate loading position thereby exposing a leading edge of a topmost printing plate in the selected plate cassette; (iv) a plate picker adapted to contact and hold the leading edge of the topmost printing plate and transport it to the plate transport pinch rollers in a first direction and orientation along the plate transport path while maintaining said orientation; (v) a paper picking unit including; (a) a paper picker adapted to contact and hold a leading edge of a topmost sheet of paper in the selected plate cassette and transport the sheet of paper in said first direction along a paper transport path as said plate held by said plate picker is transported to said plate transport pinch rollers, such that said paper and printing plate are transported substantially simultaneously; and (b) a plate guide positioned above the paper picker, said plate guide forming a part of said plate transport path, wherein the printing plate held by said plate picker is transported over said plate guide to said plate transport pinch rollers, said plate transport path being located on a first side of said plate guide and said paper transport path being located on a second side of said plate guide opposite to said first side, wherein said plate held by said plate picker is transported along said first side of said plate guide as said paper held by said paper picker is transported along said second side of said plate guide in said first direction; and (vi) an alignment system adapted to move the array of plate cassettes or the plate picker and the paper picking unit in a vertical direction to align the plate picker and the paper picking unit with the selected plate cassette.

9. The printing plate mechanism according to claim 8, wherein the plate cassettes in said array of plate cassettes are vertically aligned in a stack and said cassette movement mechanism is adapted to move the selected plate cassette in a horizontal direction to said printing plate loading position to expose the leading edge of the topmost plate.

10. The printing plate mechanism according to claim 8, wherein said paper picking unit further includes at least one pair of opposing rollers, said plate picker being adapted to transport a sheet of

paper along said paper transport path to said at least one pair of opposing rollers and said at least one pair of opposing rollers being adapted to pull the sheet of paper into said paper disposal unit.